

First-Order Trotter Difficulty for Structured and Random Pauli-String Hamiltonians

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Abstract

Product-formula Hamiltonian simulation is usually introduced through worst-case error bounds, but those bounds can be much more pessimistic than the behavior one actually sees for concrete Hamiltonians and input states. In this project, we study the practical difficulty of first-order Trotterization for several families of Pauli-string Hamiltonians using exact small-system simulation together with a separate resource-estimation pipeline. The main question is whether the number of Trotter steps needed to reach a target fidelity depends on the initial product state and its energy, and how that behavior changes across different Hamiltonian families.

This paper compares three families: a structured dipolar-inspired long-range Hamiltonian, a random two-local Pauli baseline, and a shifted local transverse-field Ising model (TFIM) control family chosen to provide a clean comparison. For small system size $N = 6$, I compute the exact evolved state $e^{-iHt} |\psi_0\rangle$, compare it with first-order Trotter evolution, and define an empirical threshold.

We also used three checks: a sweep of the error threshold ε confirming that the family ordering is preserved across accuracy targets, a comparison of empirical thresholds under different term orderings showing that r_* is not sensitive to term order, and an energy-variance diagnostic that provides a state-level predictor of Trotter difficulty complementary to the initial energy.

Overall, the results show that the practical difficulty of first-order Trotterization depends strongly on both Hamiltonian structure and initial-state choice, and that conservative worst-case step-count estimates can overshoot the observed thresholds by very large factors. When combined with the resource-estimation pipeline, the exact study also shows how those differences in Trotter difficulty translate into meaningful differences in circuit cost.

1 Methods and Setup

1.1 Exact fidelity metric and empirical threshold

For the data presented, we fixed

$$N = 6, \quad t = 1.0, \quad \varepsilon = 10^{-3}. \quad (1)$$

For each Hamiltonian instance and each initial state, the exact evolved state is computed as

$$|\psi_{\text{exact}}(t)\rangle = e^{-iHt} |\psi_0\rangle \quad (2)$$

And compared with the first-order Trotter approximation. The error metric is the state-fidelity error

$$\text{err}(r) = 1 - |\langle \psi_{\text{exact}}(t) | \psi_{\text{trot}}(t; r) \rangle|^2. \quad (3)$$

We define the empirical threshold

$$r_\star(H, \psi_0, t, \varepsilon) = \min \{r : \text{err}(r) \leq \varepsilon\}. \quad (4)$$

Thus $r_\star$ is the smallest Trotter step count that reaches the target error for the chosen Hamiltonian, state, time, and tolerance.

1.2 Scan range

The reported run uses the scan range

$$1 \leq r \leq r_{\max}, \quad r_{\max} = 500. \quad (5)$$

The workflow was built so that states failing to meet the target within the scan range could be tracked (dubbed “censored” r ’s). For a run using the fixed values above, however, every sampled state reached the target before the scan ceiling. The general handling of censored data is described in Section 1.4.

1.3 Hamiltonian families

Structured dipolar family. The structured long-range family is

$$H_{\text{dip}} = \sum_{i < j} J_{ij} (X_i X_j + Y_i Y_j - 2Z_i Z_j), \quad J_{ij} = \frac{J_0}{|i - j|^\alpha}.$$

Random two-local baseline. The random baseline is a matched two-local Pauli-string Hamiltonian

$$H_{\text{rand}} = \sum_{\ell=1}^M c_\ell P_\ell, \quad (6)$$

where each P_ℓ is drawn from the full pool of two-local Pauli strings on N qubits, i.e., operators of the form $Q_i \otimes R_j$ with $Q, R \in \{X, Y, Z\}$ and $i < j$. For N qubits there are $9 \binom{N}{2}$ such strings; for $N = 6$ this gives 135 distinct two-local operators. Strings are sampled without replacement whenever M does not exceed the pool size, so that every term in a single instance corresponds to a distinct Pauli operator.

The coefficients c_ℓ are drawn independently from a standard normal distribution $\mathcal{N}(0, 1)$ and then globally rescaled so that

$$\|c\|_2 \equiv \left(\sum_{\ell} c_\ell^2 \right)^{1/2}$$

matches the ℓ_2 norm of the structured dipolar instance being compared. This matching ensures that both families share the same overall coupling scale, so that observed differences in the empirical threshold reflect structural properties of the Hamiltonian (i.e., locality pattern, commutation structure, positional correlations) rather than a simple amplitude difference. Every random instance can be generated from a fixed integer seed for exact reproducibility; the seed, coefficient norm, sampling mode, and Pauli pool size are all recorded in the study metadata.

Shifted local TFIM control family. The local control family is

$$H_{\text{TFIM}} = J \sum_{i=1}^{N-1} \frac{I - Z_i Z_{i+1}}{2} + h \sum_{i=1}^N \frac{I - X_i}{2}.$$

1.4 Censored threshold scanning and summary statistics

For any state-instance pair that does not satisfy $\text{err}(r) \leq \varepsilon$ for any r in the scan range $[1, r_{\max}]$, the threshold $r_{\star}$ is said to be unresolved or censored. Unresolved cases are tracked explicitly: the data record carries a Boolean **resolved** flag together with the best error achieved within the scan window. Two summary statistics are computed for each family sweep.

1. $\overline{r}_{\star}^{\text{res}}$: the mean threshold computed over resolved states only.
2. $\overline{r}_{\star}^{\text{lb}}$: a lower-bound mean in which each unresolved state contributes $r_{\max}$ to the average. Because every unresolved threshold is at least $r_{\max}$, this quantity is a guaranteed floor on the true mean:

$$\overline{r}_{\star}^{\text{lb}} \leq \overline{r}_{\star}^{\text{true}}.$$

In the primary $N = 6$ run with $r_{\max} = 500$, all sampled states resolve within the scan window and the two statistics coincide. At the $N = 8$ follow-up with $r_{\max} = 300$, there are unresolved cases and thus the lower-bound mean becomes a meaningful conservative estimate distinct from the resolved-only mean. The distinction is important whenever comparing families with different resolution rates, since reporting only resolved-state averages can systematically understate the difficulty of families where many states are censored.

2 Exact Small- N Results

2.1 Family-level summary

Table 1: Family-level summary of empirical threshold behavior from the final dataset.

Family	Instances	States	Reached target	Average $r_{\star}$	Range of $r_{\star}$
Structured dipolar	1	21	21	182.62	1–253
Local shifted TFIM	1	21	21	17.29	15–19
Random two-local	15	315	315	134.43	108–162

2.2 Local shifted TFIM

The clearest correlation in the present study appears in the shifted local TFIM family. Because the threshold range is so narrow, this family gives a relatively simple picture of how the required first-order Trotter step count changes across the sampled product-state manifold.

The correlation between $r_{\star}$ and initial energy is strong and negative:

$$\text{corr}(r_{\star}, \Delta) \approx -0.964. \tag{7}$$

Within this family and at the operating point used here, states with larger Δ tend to require slightly fewer Trotter steps. The total variation is modest, since $r_{\star}$ only ranges from 15 to 19, but the trend is consistent across all 21 sampled states.

This family is also the least expensive one in the current resource model, both per Trotter step and in total compiled cost.

2.3 Structured dipolar family

The structured dipolar family shows the strongest state dependence in the exact dataset. The endpoint product states, $\theta = 0$ and $\theta = \pi$, meet the target immediately with $r_\star = 1$, while states closer to the middle of the sampled product-state path require well over 200 Trotter steps.

The correlation between $r_\star$ and initial energy is strongly positive:

$$\text{corr}(r_\star, \Delta) \approx 0.857. \quad (8)$$

For this structured dipolar Hamiltonian and this product-state family, the number of first-order Trotter steps required depends strongly on the chosen initial state.

2.4 Random two-local

Early runs made it clear that the random two-local family is not well represented by a single instance. That is why this work moved to multiple random realizations, although one instance is still kept as a baseline example for the direct three-family comparison.

In the final run, the random two-local family still shows substantial instance-to-instance variation. The direct comparison instance used alongside the structured dipolar and local TFIM families has mean

$$\bar{r}_\star \approx 137.29 \quad (9)$$

with thresholds in the range 122–162.

So the random baseline is better interpreted as a distribution over instances than as a single representative Hamiltonian.

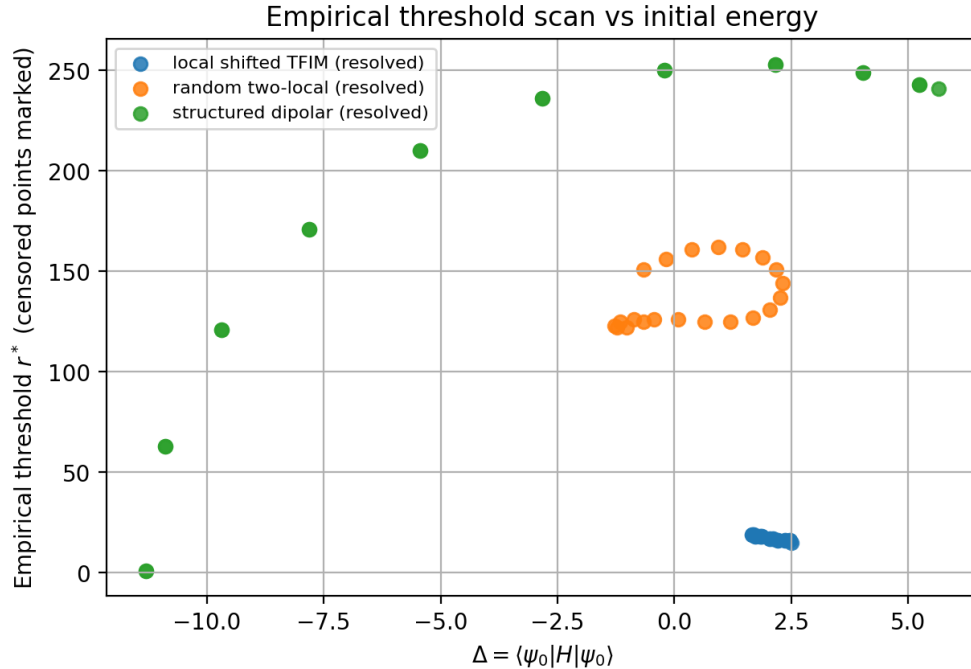


Figure 1: Empirical first-order Trotter thresholds $r_\star$ versus initial energy Δ for the exact $N = 6$ study.

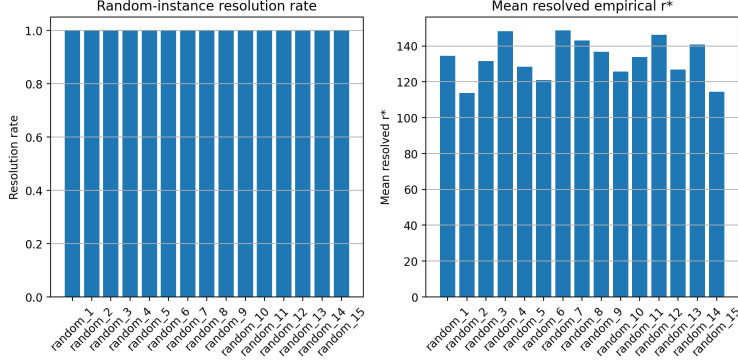


Figure 2: Instance-to-instance variation in the random two-local family. The instance `random_0` (unlabeled) provides the baseline data used in Fig. 1.

Table 2: Representative exact-state results from the final reported run.

Family	Instance	θ	Δ	$r_\star$	Error at $r_{\max}$
Structured dipolar	structured_0	0	-11.3007	1	1.14×10^{-13}
Structured dipolar	structured_0	1.0996	2.1566	253	2.54×10^{-4}
Local shifted TFIM	local_0	0.7854	1.8651	18	1.18×10^{-6}
Random two-local	random_0	0.4712	0.9367	162	1.05×10^{-4}

2.5 Energy variance diagnostic

For each Hamiltonian-state pair (H, ψ_0) we record the energy variance of the initial state,

$$\sigma_H^2(\psi_0) = \langle H^2 \rangle_{\psi_0} - \langle H \rangle_{\psi_0}^2, \quad (10)$$

computed exactly from the same matrix representation used for the Trotter evolution. Figure 3 plots $r_\star$ against σ_H^2 for all resolved states, with endpoint product states ($\theta \approx 0$ and $\theta \approx \pi$) having unique markers.

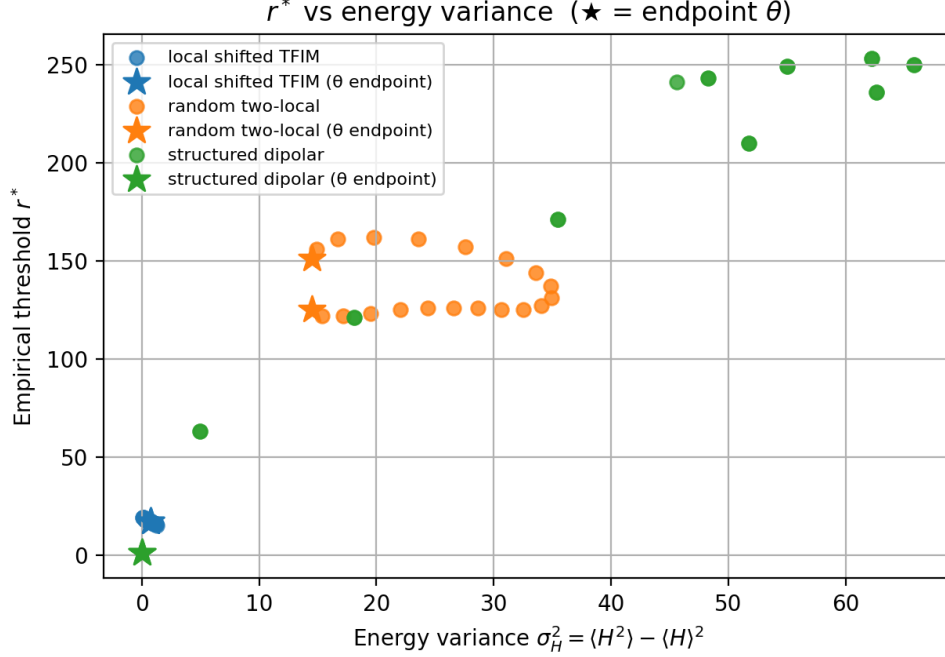


Figure 3: Empirical threshold r_* versus initial-state energy variance $\sigma_H^2 = \langle H^2 \rangle - \langle H \rangle^2$ for the exact $N = 6$ dataset. Stars mark endpoint product states ($\theta \approx 0, \pi$).

The energy variance is useful when compared to the initial energy Δ alone. A state can have large Δ while still being a near-eigenstate of H ; in that case the variance is small and the dynamics can track closely on a coarse time grid. The clearest illustration appears in the structured dipolar family: the product states at $\theta = 0$ and $\theta = \pi$ are exact eigenstates of H_{dip} , so $\sigma_H^2 = 0$ and Trotter evolution is exact for any $r \geq 1$. Interior states have nonzero energy variance and require substantially more Trotter steps, and the correlation between σ_H^2 and r_* within the structured family is positive and strong. For the local shifted TFIM family, σ_H^2 varies only weakly across the product-state path, consistent with the narrow observed range of thresholds. For the random two-local family, the variance and threshold vary from instance to instance in a way that reflects the random realization of the coupling structure, but the positive trend between variance and threshold is preserved within each instance.

3 Robustness and Sensitivity Checks

3.1 Validation: epsilon sweep

To test whether the family ordering observed at $\varepsilon = 10^{-3}$ is robust, we repeat the threshold scan for each exact-regime family at

$$\varepsilon \in \{10^{-2}, 10^{-3}\}, \quad (11)$$

holding all other parameters fixed at the values in Eq. (1). For each value of ε and each family, we compute both the mean resolved threshold $\bar{r}_*^{\text{res}}$ and the lower-bound mean $\bar{r}_*^{\text{lb}}$ described in Section 1.4.

Figure 4 shows the results. The family ordering; local shifted TFIM easiest, structured dipolar hardest, random two-local intermediate, is preserved across all three values of ε . The mean threshold increases monotonically as ε decreases for every family, as expected: tightening the accuracy target

uniformly increases the number of steps required. The spacing between families also widens at smaller ε , reflecting the fact that the error convergence rates differ across families. In the $N = 6$ run all states are resolved at every threshold and the two mean statistics coincide; the lower-bound mean becomes a distinct, more conservative estimate at larger N or smaller $r_{\max}$.

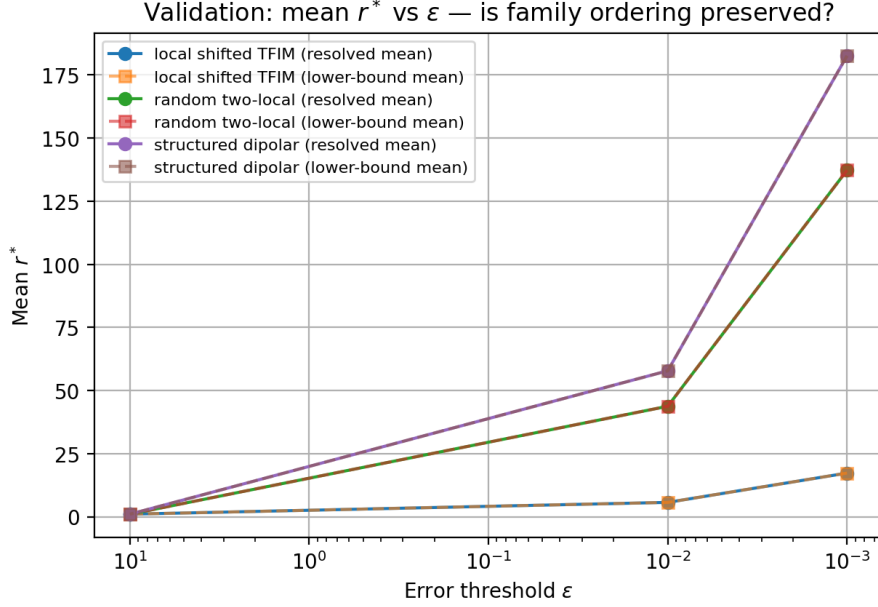


Figure 4: Mean empirical threshold $r_{\star}$ versus error tolerance ε for each family. Solid markers show resolved-only means; dashed markers show lower-bound means (coincident here because all states are resolved within the scan range).

3.2 Trotter ordering sensitivity

The first-order product formula depends on the ordering of the Hamiltonian terms. Reordering changes the specific commutator error terms at each step, and while the leading-order error bound is insensitive to ordering, the empirical threshold $r_{\star}$ could in principle shift. To assess this, we compare $r_{\star}$ under four orderings of the structured dipolar terms: the default forward ordering, the reversed ordering, and two fixed random permutations drawn independently with the study seed. Each ordering defines a fixed term sequence applied identically at every Trotter step.

Figure 5 shows box plots of the resolved $r_{\star}$ values under each ordering across the 21-state product-state family. The distributions overlap substantially: the interquartile ranges and medians are similar across all four orderings, and no ordering is systematically easier or harder. We conclude that the empirical threshold is not strongly sensitive to term ordering at this operating point and system size, which gives additional confidence that the family-level comparisons are not artifacts of a particular term sequence.

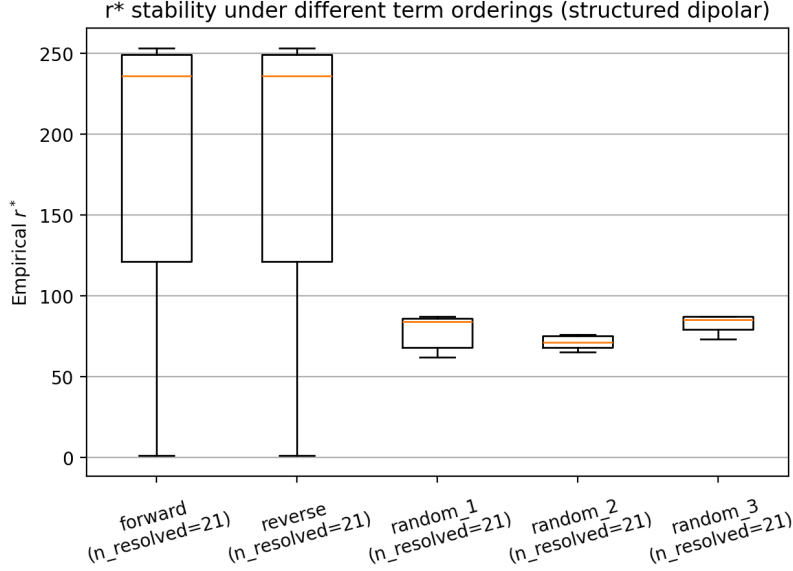


Figure 5: Distribution of empirical thresholds r_* for the structured dipolar family under forward, reverse, and two fixed random term orderings. Each box spans the interquartile range of resolved r_* values across the 21-state product-state family.

4 Comparison with Worst-Case Step Estimates

The current workflow includes a commutator-based worst-case estimate of the form

$$\frac{t^2}{2r} \sum_{a < b} \|[H_a, H_b]\| \leq \varepsilon. \quad (12)$$

This gives a rough, state-independent estimate for the number of Trotter steps required for each Hamiltonian instance. It is not used as a rigorous bound for the exact fidelity error in Eq. (3), rather a conservative comparison point.

Even with that caveat, the estimate is still useful as a diagnostic tool. For the local shifted TFIM instance, it recommends $r \approx 1751$, while the exact thresholds are only 15–19. For the structured dipolar instance, the estimate is approximately 63114, while the exact thresholds range from 1 to 253. For the direct random-comparison instance, the estimate is approximately 184566, while the exact thresholds range from 122 to 162.

The conclusion is not that the worst-case estimate is formally wrong. Rather that, in this small- N setting, it is extremely conservative relative to the thresholds actually observed. In the final run, the ratio between the worst-case estimate and the observed threshold is roughly 10^2 for the local TFIM family, roughly 10^3 for the random baseline, and between about 2.5×10^2 and 6.3×10^4 for the structured dipolar family depending on the initial state.

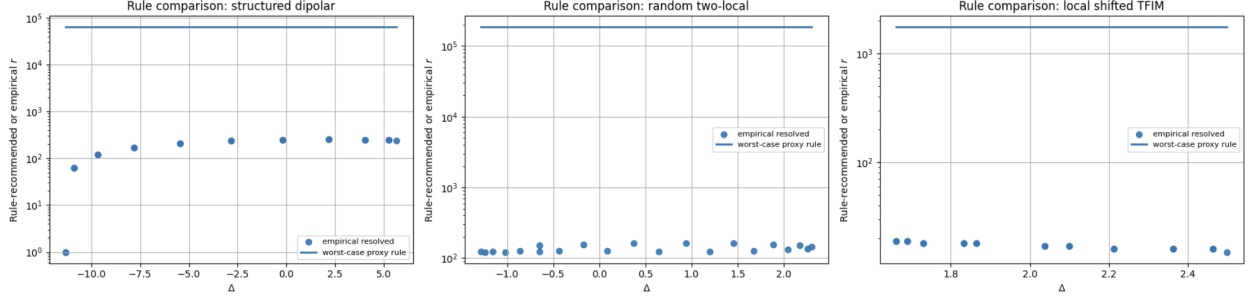


Figure 6: Comparison between exact empirical thresholds and worst-case step estimates.

Table 3: Representative comparison between observed thresholds and conservative worst-case estimates.

Family / instance	Typical empirical $r_\star$	Worst-case estimate r	Ratio
Structured dipolar	1–253	63114	$\sim 2.5 \times 10^2$ to 6.3×10^4
Local shifted TFIM	15–19	1751	$\sim 10^2$
Random two-local (random_0)	122–162	184566	$\sim 10^3$

5 Full-Stack Impact / Resource Implications

A natural next question is how the differences seen in the exact study translate into circuit cost. The direct connection is that smaller empirical thresholds mean fewer repeated Trotter steps, which in turn means lower estimated gate counts and lower non-Clifford overhead. In the present workflow, the clearest cost quantity is the estimated T-count.

At the main exact-study size $N = 6$, the per-step resource model already separates the Hamiltonian families clearly. The structured dipolar and random two-local families each contain 45 non-identity two-site terms and require about 90 CNOTs per first-order Trotter step, with estimated T-counts of 4526 and 4528, respectively. By contrast, the shifted local TFIM family contains only 11 nonidentity terms total, of which 5 are two-site terms, and requires only 10 CNOTs and an estimated T-count of 1100 per step. Once the empirical thresholds are folded in, this gap becomes much larger. Averaged over states that meet the target error, the mean total estimated T-count is approximately 1.92×10^4 for the local shifted TFIM family, 6.10×10^5 for the pooled random two-local family, and 8.26×10^5 for the structured dipolar family. In other words, the families that are harder in the exact fidelity study are also much more expensive in total non-Clifford cost.

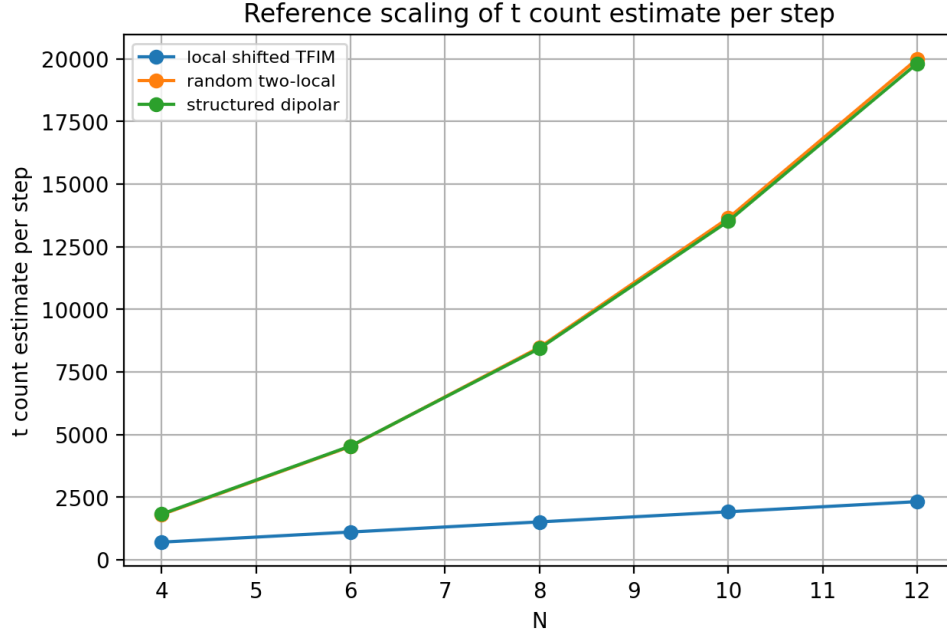


Figure 7: Per-step resource estimates under the chosen compilation model. We can see slight splitting of random two-local and structured dipolar at relatively small $N=12$, so if ordering is preserved, there would be a significant difference at much larger N

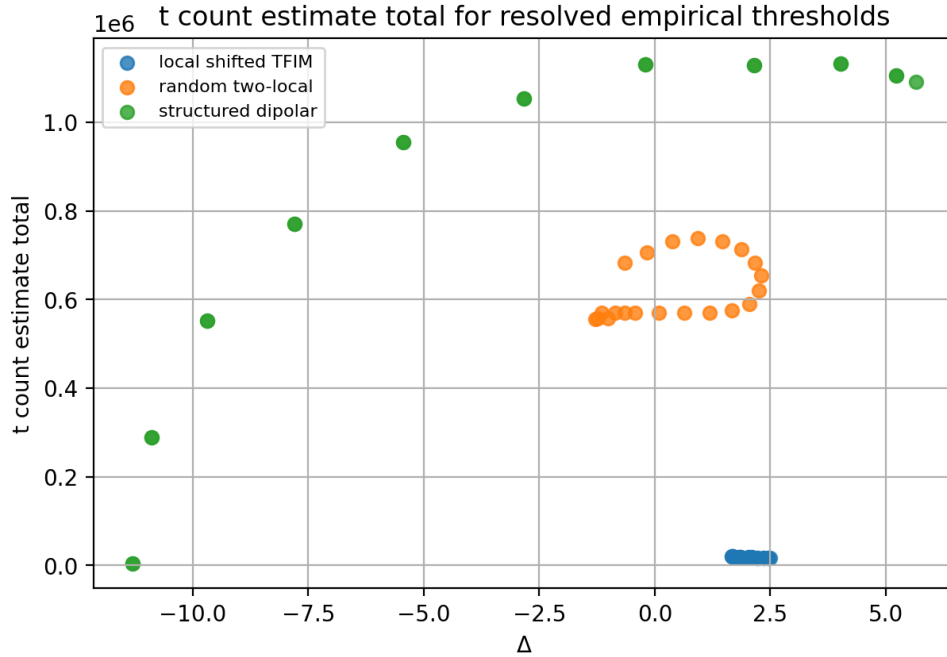


Figure 8: Total estimated T-count for states meeting the target error, using the empirical thresholds.

To check whether this picture survives at a slightly larger exact size, we also ran a follow-up

exact study at

$$N = 8, \quad t = 1.0, \quad \varepsilon = 10^{-3}, \quad r_{\max} = 300,$$

using the same 21-state product-state family and the same three Hamiltonian classes. The broad family ordering is preserved. The shifted local TFIM family remains the easiest case, with all 21 sampled states reaching the target and

$$r_{\star} \in [18, 23], \quad \overline{r}_{\star} \approx 20.38.$$

The random two-local family also remains fully resolved across all 15 sampled instances, with pooled thresholds

$$r_{\star} \in [142, 201], \quad \overline{r}_{\star} \approx 170.27.$$

The structured dipolar family remains the most state-sensitive: only 10 of the 21 sampled states reach the target within the scan range, and among those resolved states

$$r_{\star} \in [1, 265], \quad \overline{r}_{\star} \approx 141.0.$$

Note that for the $N = 8$ structured dipolar family, $\overline{r}_{\star}^{\text{res}}$ and $\overline{r}_{\star}^{\text{lb}}$ diverge: the resolved-only mean is 141.0, while the lower-bound mean (treating the 11 unresolved states as $r_{\max} = 300$) is substantially larger, providing a more conservative summary of the overall difficulty.

The corresponding resource costs also grow substantially at $N = 8$. The mean total estimated T-count is approximately 3.05×10^4 for the local shifted TFIM family, 1.44×10^6 for the pooled random two-local family, and 1.20×10^6 for the resolved structured dipolar states. Even though the $N = 8$ run is smaller in scope than the main $N = 6$ study because of device limitations, it supports the same qualitative conclusion: the family that is easier in the exact fidelity study also remains much cheaper in total non-Clifford cost.

I also carried out a larger- N fixed-step resource calculation, summarized in Figure 7 and Table 5. This is not an exact fidelity study; it is included only to show how the cost of a single Trotter step grows with system size. The same conclusion is represented here: the shifted local TFIM family grows much more gently, while the structured dipolar and random two-local families remain much more expensive per step.

Table 4: Combined exact-study and follow-up resource summary. For the structured dipolar family at $N = 8$, averages are over resolved states only, the lower-bound mean is larger.

Family	N	# resolved	Avg. resolved $r_{\star}$	T-count / step	Avg. total estimated T-count
Local shifted TFIM	6	21/21	17.29	1100	1.92×10^4
Random two-local	6	315/315	134.43	4528	6.10×10^5
Structured dipolar	6	21/21	182.62	4526	8.26×10^5
Local shifted TFIM	8	21/21	20.38	1500	3.05×10^4
Random two-local	8	315/315	170.27	8506	1.44×10^6
Structured dipolar	8	10/21	141.0	8444	1.20×10^6

Table 5: Reference resource scaling from the fixed-step resource study, restricted to $N \in \{6, 8, 12\}$.

Family	N	Terms	CNOTs / step	T-count / step	Note
Local shifted TFIM	6	11	10	1100	study size
Local shifted TFIM	8	15	14	1500	follow-up size
Local shifted TFIM	12	23	22	2300	reference only
Random two-local	6	45	90	4528	study size
Random two-local	8	84	168	8506	follow-up size
Random two-local	12	198	396	20002	reference only
Structured dipolar	6	45	90	4526	study size
Structured dipolar	8	84	168	8444	follow-up size
Structured dipolar	12	198	396	19956	reference only

6 Limitations and Scope

This project is best understood as a controlled computational study of first-order Trotterization in a small-system regime with a follow-up exact study at $N = 8$ and a separate fixed-step resource analysis at larger N . Within that scope, the workflow gives a clear picture of how Trotter difficulty changes across Hamiltonian families and across initial states.

Because the initial energy Δ is not normalized across families, the cleanest interpretation is within-family rather than cross-family: the data show that initial-state choice can matter substantially. Likewise, the shifted local TFIM family differs from the dipolar and random families in several ways at once, including sparsity and locality, so this study is strongest as a comparison of practical Trotter difficulty across representative Hamiltonian classes.

The robustness checks in Section 3 address two specific concerns raised in peer review. The epsilon sweep (Section 3.1) confirms that the family ordering is not specific to the choice $\varepsilon = 10^{-3}$. The ordering comparison (Section 3.2) confirms that the empirical thresholds are not sensitive to the particular sequence in which terms are applied. Both checks were performed at $N = 6$ and are intended to strengthen confidence in the qualitative conclusions rather than to extend the scope of the study.

The larger- N scaling section should also be read as a resource study rather than an exact accuracy study. Its purpose is to show how the cost of a Trotter step grows with system size and how that growth differs across Hamiltonian families.

7 Discussion

The main result of this project is that the practical difficulty of first-order Trotterization depends strongly on Hamiltonian structure and, in many cases, on the initial state as well. That conclusion comes directly from exact computational data.

The shifted local TFIM family provides the simplest example of an easy regime: all sampled states reach the target with low step counts, and the variation across the product-state path is small but systematic. The structured dipolar family shows much stronger state dependence, with thresholds ranging from 1 to 253 along the same style of sampled path. The random two-local family falls between these cases, while also showing that instance-to-instance variation is itself part of the computational story.

The energy variance diagnostic (Section 2.5) offers a complementary perspective on why certain states are harder. For the structured dipolar family, the endpoint states are exact eigenstates of the Hamiltonian and therefore have $\sigma_H^2 = 0$ and trivial Trotter error. Interior states, which have nonzero energy variance, require substantially more steps. This suggests that σ_H^2 may be a more direct predictor of Trotter difficulty than the initial energy Δ alone, at least within a single Hamiltonian family.

A second takeaway is that conservative worst-case step estimates can differ dramatically from observed small-instance behavior. In this study, the commutator-based estimate consistently overpredicts the empirical thresholds, sometimes by orders of magnitude. That makes exact computational studies especially useful for understanding how Trotterization behaves on specific Hamiltonians that are actually being simulated.

The resource analysis points in the same direction. The Hamiltonian families that are easier in the exact fidelity study are also the ones that are cheaper once the simulation is translated into circuit-level cost. In particular, the shifted local TFIM family remains much cheaper both per step and in total estimated T-count, while the structured dipolar and random two-local families remain substantially more expensive.

Overall, the results support a computational picture in which Trotter difficulty is shaped by Hamiltonian structure, locality, commutation behavior, and initial-state choice together.

8 Conclusion

This paper presents a reproducible computational study of first-order Trotter step counts for structured and random Pauli-string Hamiltonian families. The main conclusion is that the practical difficulty of first-order Trotterization depends strongly on the Hamiltonian family and can also depend strongly on the initial state. The robustness checks confirm the qualitative family ordering is preserved across thresholds and is not sensitive to term ordering. More broadly, this study shows that exact small- N computations can reveal a more detailed picture of Trotterization than conservative worst-case estimates alone. When combined with the resource-estimation pipeline, those exact results also show how empirical differences in Trotter difficulty translate into meaningful differences in circuit cost.

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